

title: D · Value-Added Components description: Seven strategic differentiators offered at no additional cost — bilingual UI, offline-first PWA, HR Copilot, payroll anomaly detector, hash-chained audit ledger, transparency portal, 10-year escrow.

D · Value-Added Components

!!! info "About this paper" Papers **A**, **B** and **C** describe the base scope demanded by the PBD. **This paper is different** — it describes seven **discretionary components** offered *in addition* to the base scope, at **no incremental cost** to the ABC of PHP 500 M. Each one is either a re-use of a component already scoped, a hardening of a mandatory requirement, or a discharge of a risk that would otherwise fall on DepEd. Formal contract wording is in [Paper B §16.5](#).

Design principle

Every value-added component (VAC) had to pass **three gates** before being included:

1. **Documented pain point.** The PBD acknowledges the need (or the operating context makes it obvious) but does not fully price a solution.
2. **Fits inside the 365 days.** No displacement of base scope; delivery slotted into an existing milestone.
3. **Survives audit.** Passes CSC, COA, NPC, and DICT review with no additional legal or procurement action.

Ideas that did not clear all three gates are recorded but not offered — because a VAC that becomes a scope conversation is a liability, not a differentiator.

The seven components

1 · Bilingual UI + SMS / USSD channels

What. Full user interface in Filipino, Cebuano, and Ilocano in addition to English. Payslip enquiries, leave applications, and OTP verification also reachable via SMS and USSD short-codes, so any teacher with a feature phone can transact.

Why. Roughly 60% of DepEd's non-urban workforce still relies on SMS as a primary channel. Competitors typically ship English-only web/mobile.

Delivery. i18n framework at M2, translations delivered at M4, SMS/USSD gateway integrated at M4 alongside the notification service.

Base-scope tie-in. Extends the web and mobile UI scoped in §5 Architecture; the notification service already exists.

Evidence. Language coverage per module, SMS/USSD test-plan and gateway-provider certification (Annex V-01).

2 · Offline-first PWA with CRDT sync

What. Progressive Web App usable in schools with intermittent connectivity. Writes are captured locally using conflict-free replicated data types (CRDTs) and reconciled deterministically when the school reconnects. Kiosk daemon does the same for shared-device stations.

Why. The PBD requires “offline access” but does not specify the reconciliation strategy. Naive last-write-wins loses data at scale. CRDTs are proven — the same maths behind Google Docs and Apple Notes.

Delivery. PWA shell at M4; CRDT sync layer at M5; field-tested at three low-connectivity SDOs before M6 payroll gate.

Base-scope tie-in. Uses the same auth, encryption and schema as the online UI; only the transport and merge strategy differ.

Evidence. Reconciliation test-plan, chaos-test report showing zero-loss under 72-hour partition (Annex V-02).

3 · HR Copilot — self-hosted LLM

What. A conversational assistant grounded in the 201 file store, plantilla, DepEd Orders, and CSC memoranda. Answers natural-language questions such as *“How many secondary teachers in Region VIII are eligible for reclassification this fiscal year?”* Returns cited passages with the answer, never hallucinated free-text.

Why. Turns the HRIS from a database into a decision tool. The evaluation panel will visibly recognise this as a modern capability few competitors will demo credibly.

Delivery. Retrieval index at M5; assistant UI + guardrails at M7. Runs on a **self-hosted, open-weights model** (Llama-3.1-8B-Instruct or Qwen-2.5-7B-Instruct class) inside the DepEd DMZ — no data leaves the perimeter, satisfying RA 10173 by construction.

Base-scope tie-in. Uses the search/retrieval engine already scoped for document management and reporting.

Evidence. Grounding evaluation set with citation-accuracy scores, DPO opinion on data-flow topology, red-team results (Annex V-03).

4 · Payroll anomaly detector — M6 parallel-run gate

What. Statistical + rules-based anomaly detector that runs against every payroll cycle **before** disbursement is authorised. Flags: ghost employees, duplicate bank accounts, out-of-band net-pay deltas, deduction imbalances, GSIS/BIR/PhilHealth remittance drift, and time-clock anomalies.

Why. Turns the mandatory M6 parallel-run from a compliance chore into a **fraud-prevention asset**. Directly reduces DepEd’s post-audit exposure and the personal risk of the disbursing officer.

Delivery. Detector at M4 (training on legacy payroll data); gated integration at M6 alongside the parallel-run sign-off; deployed in production at M8.

Base-scope tie-in. Analytical extension of the payroll engine scoped in Paper C. Rules and thresholds are configurable by DepEd finance staff.

Evidence. Detection precision/recall on historical DepEd payroll (with anonymisation), operator runbook, escalation matrix (Annex V-04).

5 · Immutable, hash-chained audit ledger

What. Every state transition (personnel event, approval, payroll adjustment, permission change) is written to an append-only audit table whose rows are cryptographically hashed into a chain. A daily Merkle root is published to a public timestamp service (e.g. `opentimestamps.org` anchored to Bitcoin block headers).

Why. Blockchain-grade tamper-evidence at PostgreSQL cost. Any COA or Ombudsman investigator can independently verify that a given audit record existed on a given date and has not been altered.

Delivery. Schema at M2; hash-chain writer at M5; public timestamp job at M5; verification CLI shipped in the operations toolkit.

Base-scope tie-in. Extends the audit-log requirement in §7 Security. No new database, no new dependency — just a small extension of the existing audit schema plus a nightly cron.

Evidence. Verification demo, cryptographic protocol document, NPC-approved public-anchor plan (Annex V-05).

6 · Public transparency portal

What. A read-only, aggregate, anonymised public site: teacher-to-student ratios per region, plantilla vacancies by division, gender/age pyramid, average time-to-fill for CSC-mandated posts. No personally identifiable information ever leaves the perimeter — the portal is generated from a nightly snapshot with $k\text{-anonymity} \geq 5$ applied to every cell.

Why. DepEd looks transparent, media get a self-serve source, civil-society researchers stop filing FOI requests for the same aggregates. Zero PII risk under RA 10173.

Delivery. Snapshot pipeline at M6; portal UI at M7; NPC-approved DPIA at M6.

Base-scope tie-in. Reuses the reporting engine, the same charts library, and the existing static-site build. No new authentication or infrastructure.

Evidence. DPIA report, $k\text{-anonymity}$ certification, sample datasets (Annex V-06).

7 · 10-year source escrow + community edition

What. Two complementary continuity commitments beyond what the PBD minimum requires:

- **10-year source escrow** with a Philippine escrow agent — DepEd receives release triggers on bidder insolvency, missed critical patches, or agreed non-performance conditions.
- **Community edition** — a stripped-down variant of the HRIS core released under a permissive open-source licence, so other national agencies and LGUs can fork, contribute, or seed their own HRIS efforts.

Why. Directly answers the perennial procurement worry: *“What happens if the vendor disappears?”* Also positions DepEd as a leader in the government-tech open-source movement (aligning with DICT and DBM-PS policy direction).

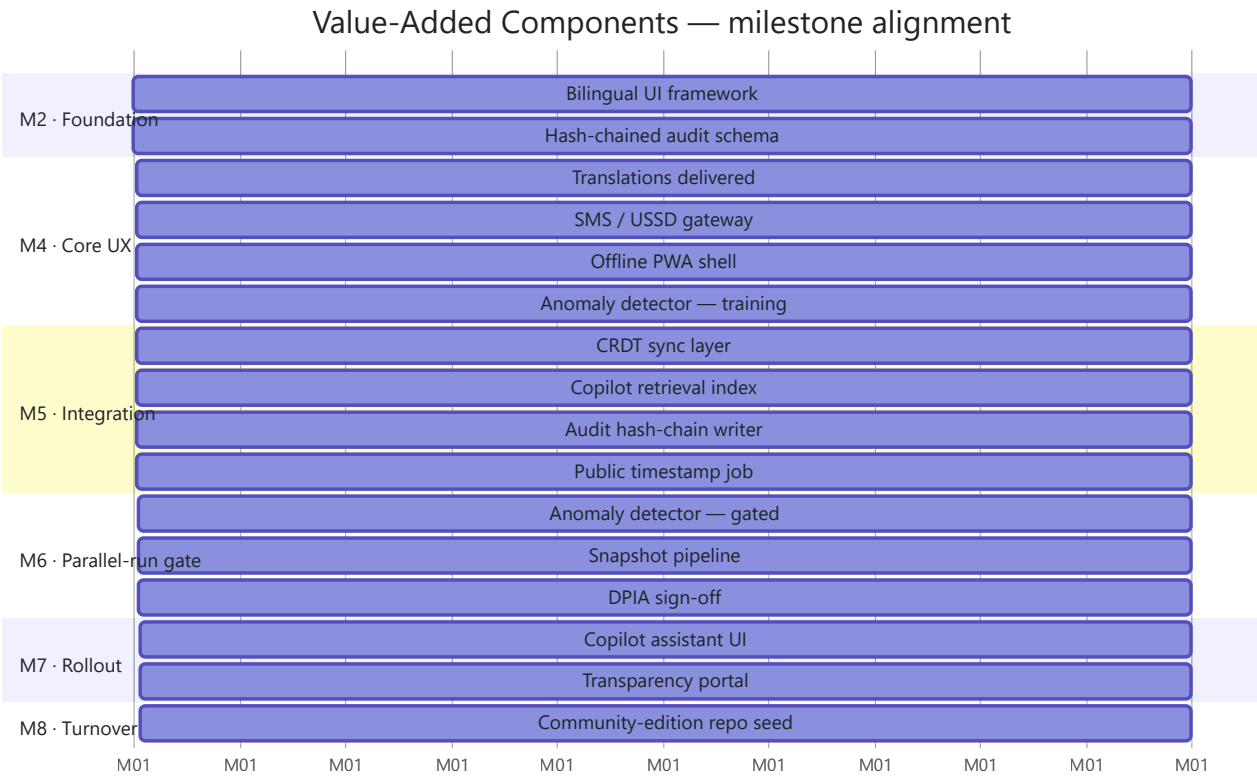
Delivery. Escrow agreement executed at M1; community edition repository seeded at M8; ongoing quarterly public releases.

Base-scope tie-in. Extends the §11 Turnover commitments. No additional engineering cost beyond community-edition curation.

Evidence. Executed escrow deed, licence text, governance charter for the community project (Annex V-07).

Delivery timeline

The seven VACs are folded into the base milestone plan — no additional milestones, no critical-path additions:



Fit against the compare table

VAC	Reinforces paper
Bilingual UI + SMS/USSD	A (scope) · C (UI/notification service)
Offline PWA + CRDT	A (offline) · C (Container view)
HR Copilot	C (retrieval component) · B (SLA)
Payroll anomaly detector	A (M6 gate) · B (milestone entry/exit)
Hash-chained audit ledger	A (audit) · B (§7 Security response)
Public transparency portal	A (28 reports) · C (reporting engine)
10-year escrow + community edition	A (turnover) · B (§11 Continuity)

Formal offer wording

The formal, contractual wording of these VACs — including the “goodwill reduction” clause and delivery gating — is in [Paper B §16.5](#).

Related: [A](#) · [Brief](#) · [B](#) · [Response](#) · [C](#) · [Architecture](#) · [Downloads](#)